Skill - 1

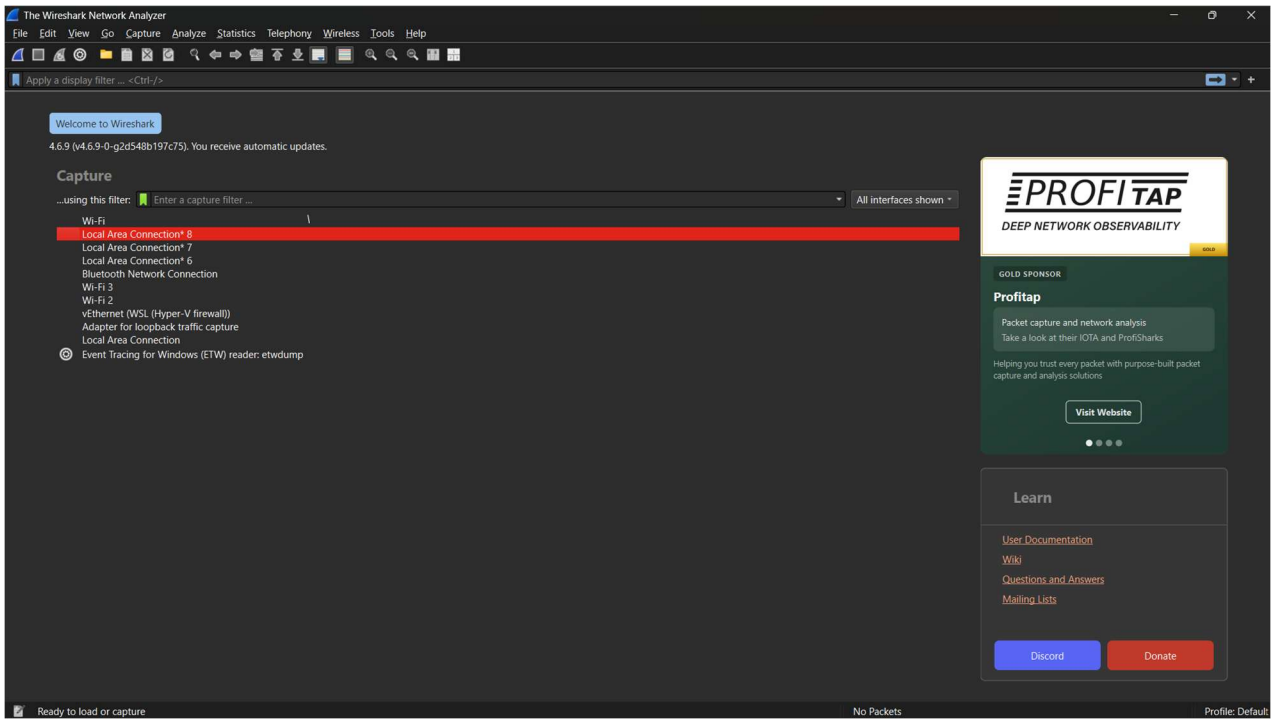
Encapsulation Builder: Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission.

Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead. Using Wireshark, we will capture real network packets, examine Ethernet, IP and TCP/UDP headers, calculate the total overhead, and determine how efficiently the network carries application data.

Step 1: Install Wireshark

1. Open the browser.
2. Install Wireshark [Wireshark • Go Deep | Download]
3. Install it using the default settings.
4. Open Wireshark.

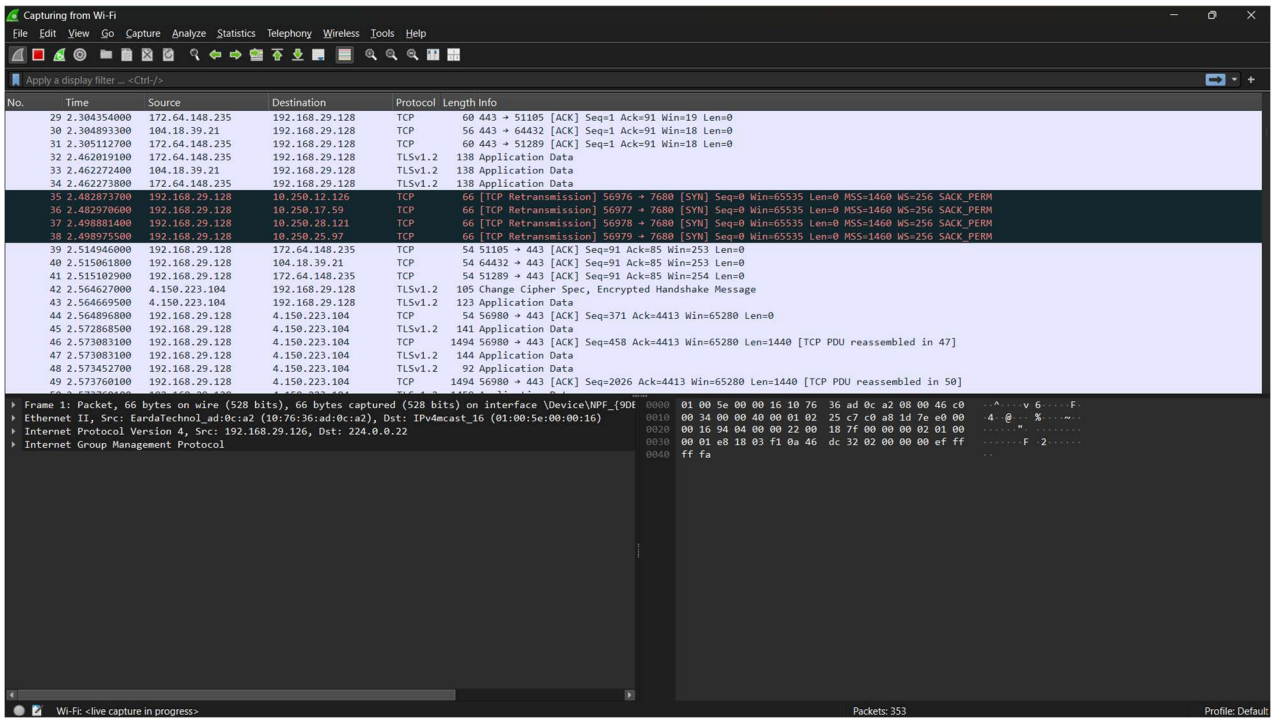


Step 2: Select your network interface

When Wireshark opens, you will see interfaces such as:

Wi-Fi, Ethernet, Bluetooth

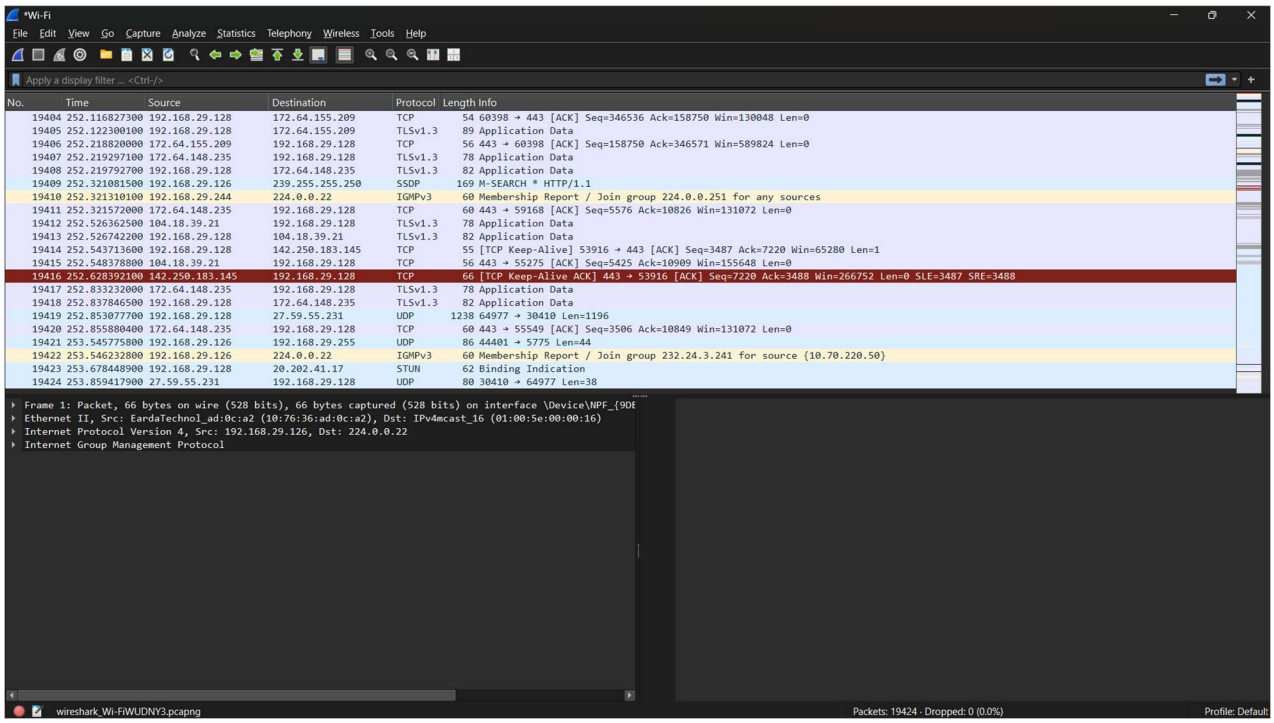
1. If you are connected through Wi-Fi, select Wi-Fi.
2. Double-click the Wi-Fi interface.
3. Packet capture will start.



Step 3: Generate network traffic While Wireshark is capturing:

1. Open a new browser tab.
2. Visit a website such as https://www.google.com.
3. Open two or three different websites.
4. Wait for approximately 30 seconds.
5. Return to Wireshark.
6. Click the red Stop button.

You should now see captured packets.

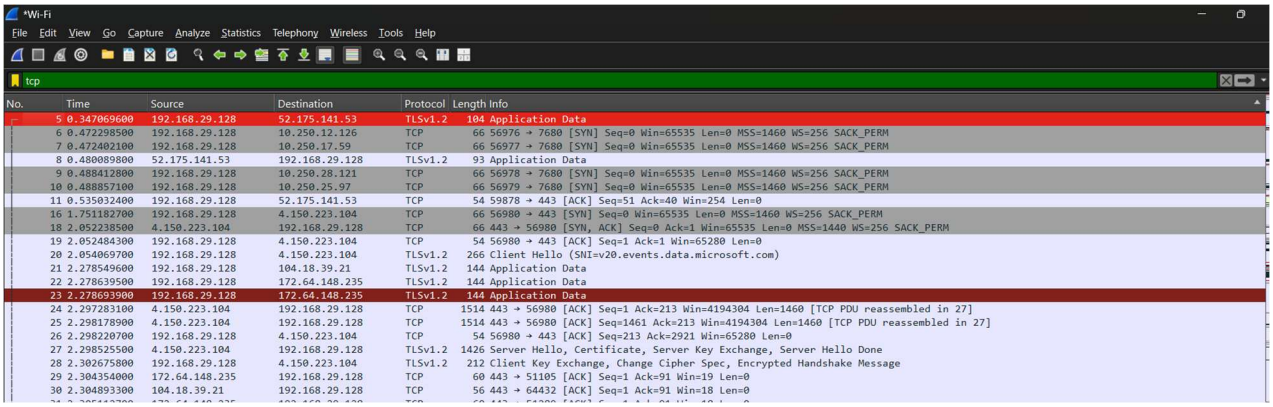


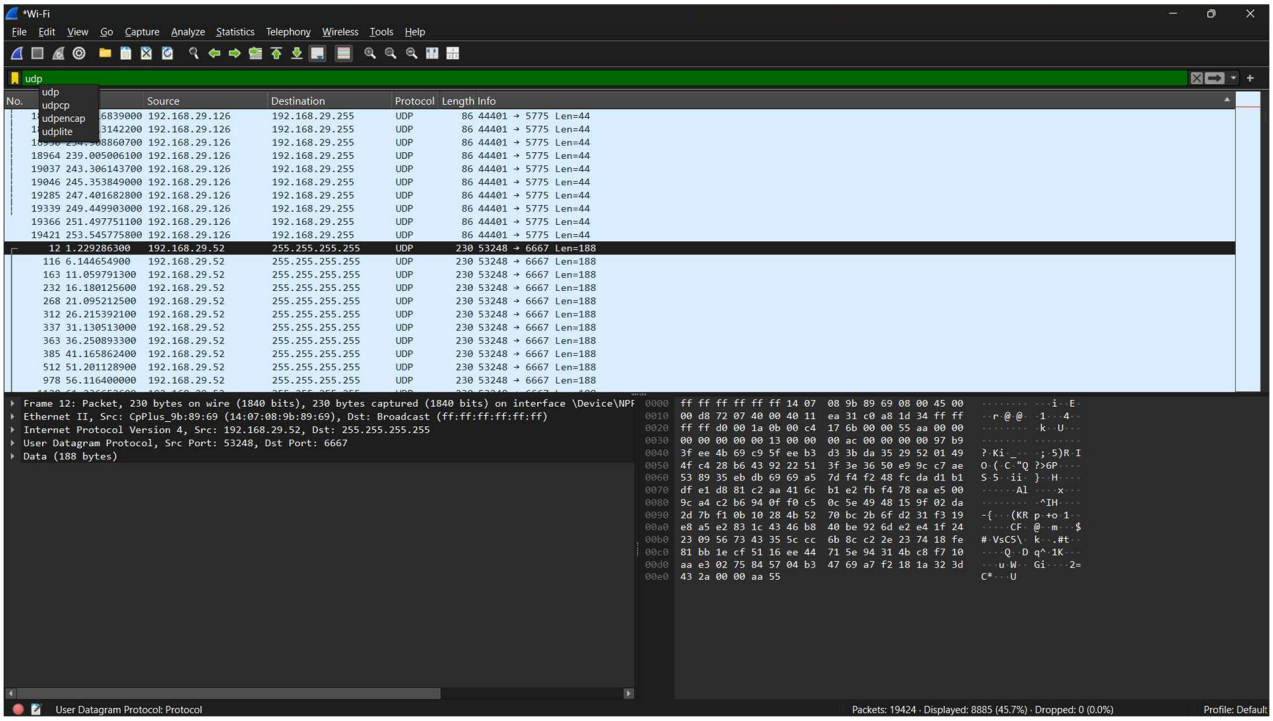
Step 4: Apply display filters
In the Wireshark filter bar, enter each filter separately and press Enter.

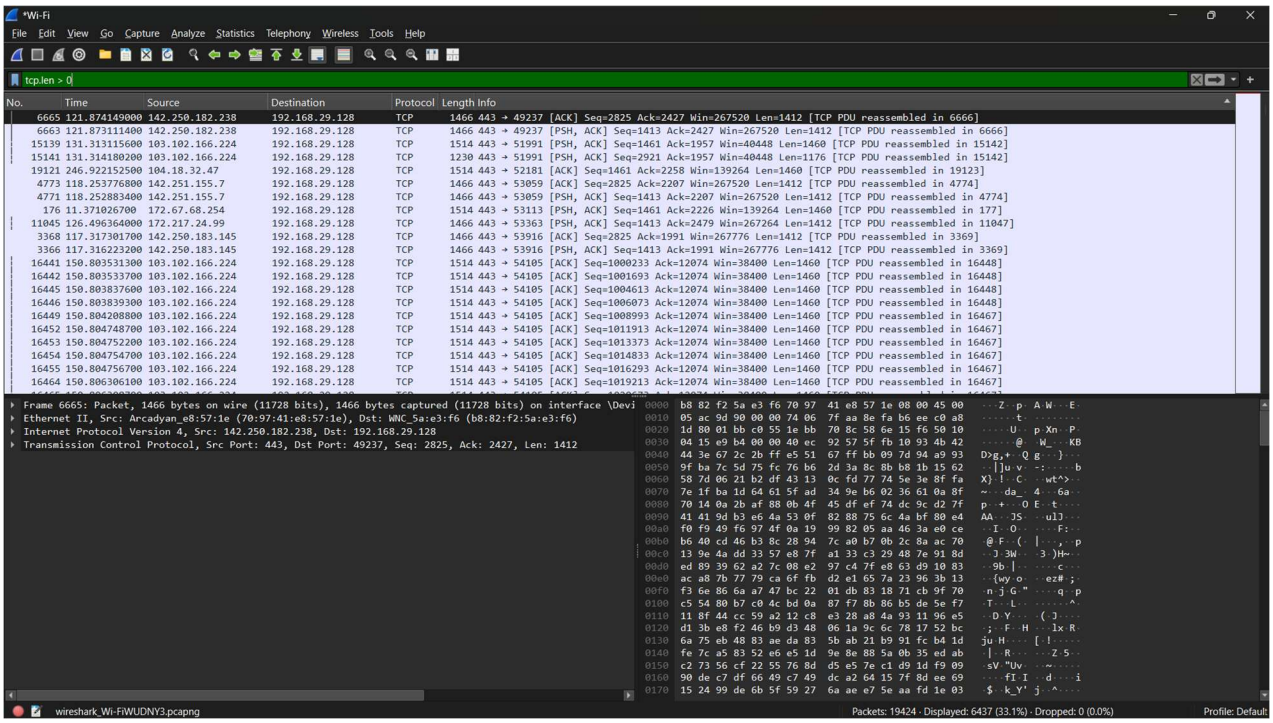
Filter	**Purpose**
tcp	Displays TCP packets
udp	Displays UDP packets
icmp	Displays ICMP packets
tcp.len > 0	Displays TCP packets carrying data

First, enter: TCP
Observe the packets displayed.
Then enter: UDP

Observe the UDP packets.





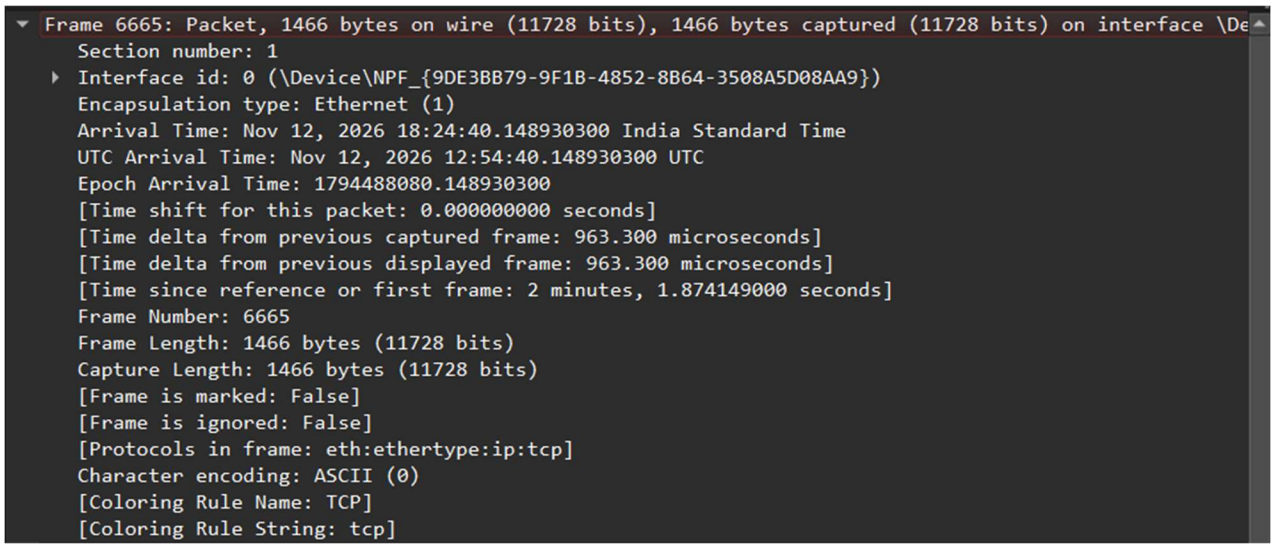


Step 5: Inspect packet headers

Select any TCP packet from the packet list.

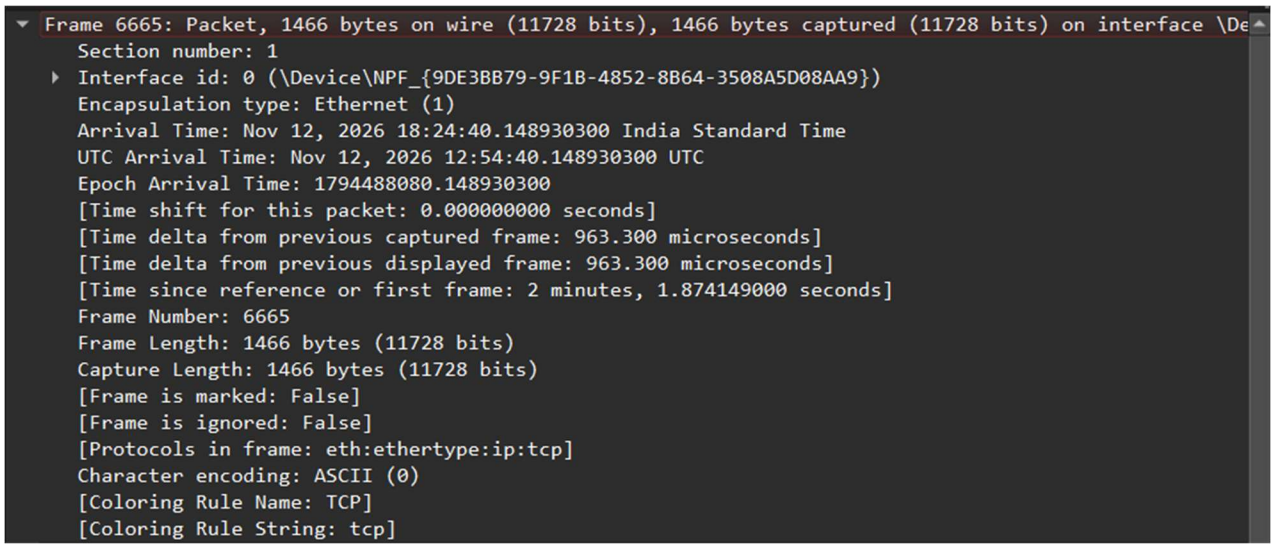
In the packet details pane, expand these sections:

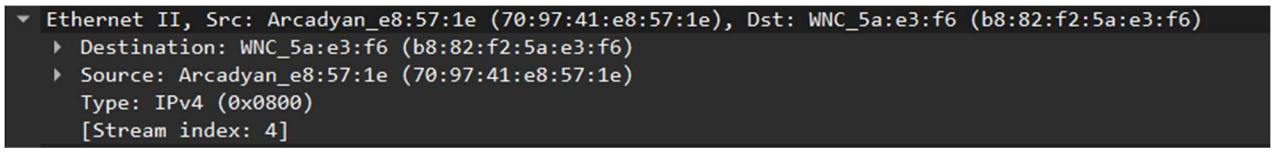
1. Frame
2. Ethernet II
3. Internet Protocol Version 4
4. Transmission Control Protocol

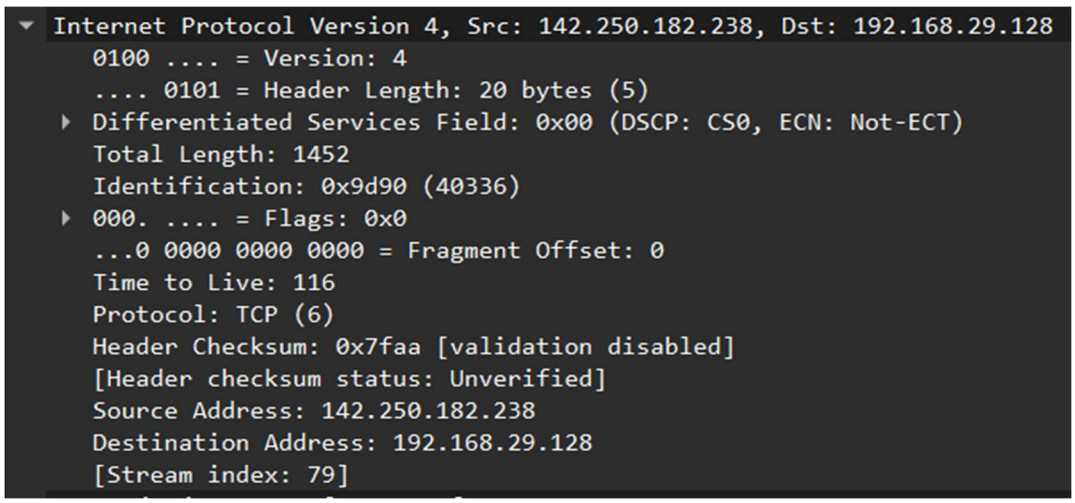


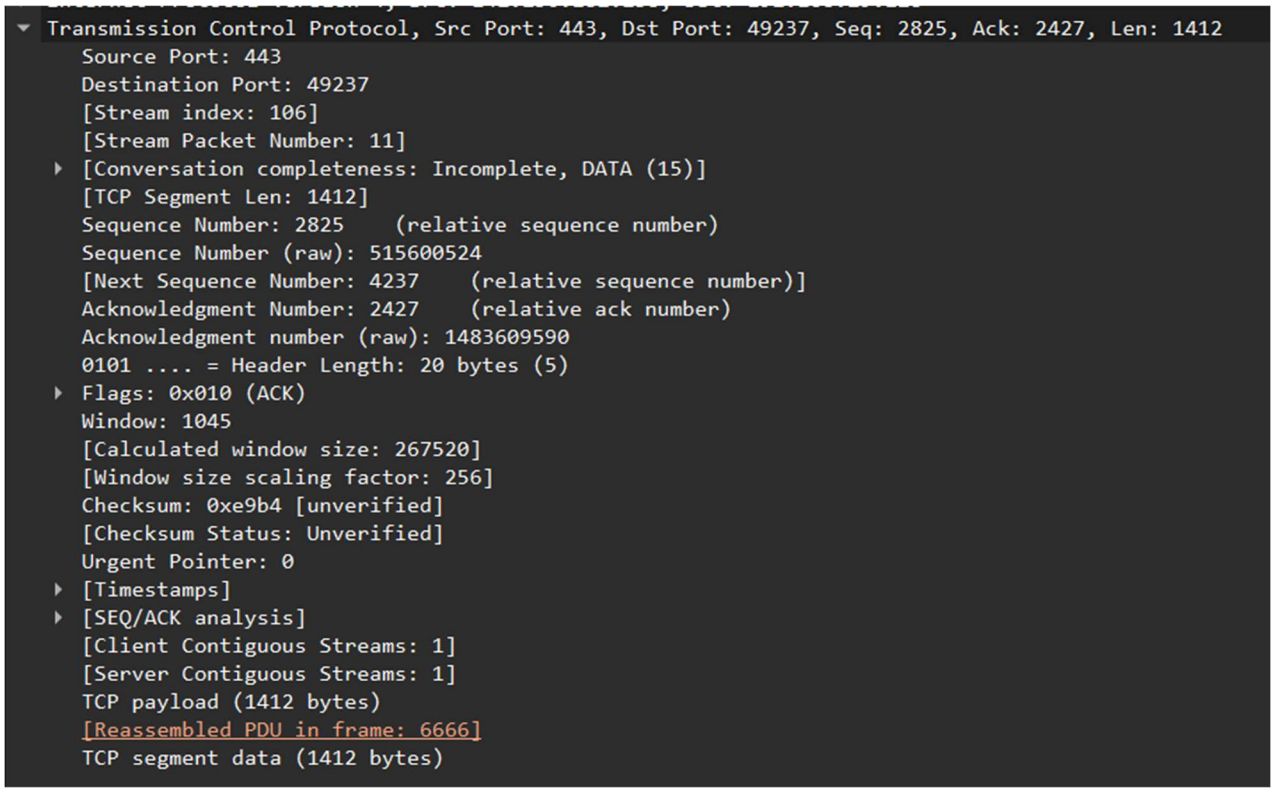
Record the following values:

Parameter	**Where to find it**
Frame length	Frame
Ethernet Header	Ethernet II
Source MAC Address	Ethernet II
Destination MAC Address	Ethernet II
IPv4 Header Length	Internet Protocol Version 4
IPv4 Total Length	Internet Protocol Version 4
TCP Header Length	Transmission Control Protocol
TCP Segment Length	Transmission Control Protocol


Frame 6665: Packet, 1466 bytes on wire (11728 bits), 1466 bytes captured (11728 bits) on interface \De
Section number: 1
Interface id: 0 (\Device\NPF_{9DE3BB79-9F1B-4852-8B64-3508A5D08AA9})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 12, 2026 18:24:40.148930300 India Standard Time
UTC Arrival Time: Nov 12, 2026 12:54:40.148930300 UTC
Epoch Arrival Time: 1794488080.148930300
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 963.300 microseconds]
[Time delta from previous displayed frame: 963.300 microseconds]
[Time since reference or first frame: 2 minutes, 1.874149000 seconds]
Frame Number: 6665
Frame Length: 1466 bytes (11728 bits)
Capture Length: 1466 bytes (11728 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]


Ethernet II, Src: Arcadyan_e8:57:1e (70:97:41:e8:57:1e), Dst: WNC_5a:e3:f6 (b8:82:f2:5a:e3:f6)
Destination: WNC_5a:e3:f6 (b8:82:f2:5a:e3:f6)
Source: Arcadyan_e8:57:1e (70:97:41:e8:57:1e)
Type: IPv4 (0x0800)
[Stream index: 4]


Internet Protocol Version 4, Src: 142.250.182.238, Dst: 192.168.29.128
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 1452
Identification: 0x9d90 (40336)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 116
Protocol: TCP (6)
Header Checksum: 0x7faa [validation disabled]
[Header checksum status: Unverified]
Source Address: 142.250.182.238
Destination Address: 192.168.29.128
[Stream index: 79]



Repeat the process for one UDP packet. Observe the UDP header and its length field.

Step 6: Calculate header overhead

For a basic Ethernet + IPv4 + TCP packet without TCP options:

- Ethernet header = 14 bytes
- IPv4 header = 20 bytes
- TCP header = 20 bytes
- Application data = 1500 bytes
- FCS = 4 bytes, if included in the frame representation

Total size including FCS:

$$14 + 20 + 20 + 1500 + 4 = 1558 bytes$$

Total overhead:

$14 + 20 + 20 + 4 = 58bytes$

Payload efficiency:

$$\frac{1500}{1558} \times 100 = 96.28\%$$

Note: Wireshark may not capture or display the Ethernet FCS. Use the actual captured packet fields when recording your observations.

Step 7:
Fill in your observation table

Record values from your own Wireshark capture.

Parameter	**TCP Packet**	**UDP Packet**
Frame Length	1466 bytes	Not provided
Ethernet Header Length	14 bytes	Not provided
IP Header Length	20 bytes	Not provided
Transport Header Length	20 bytes	Not provided
IP Total Length	1452 bytes	Not provided
Transport Header Length	1432 bytes	Not provided
Payload Length	1412 bytes	Not provided

Answers:

1. What is encapsulation?

   Encapsulation is the process of adding protocol headers to data at each network layer before transmitting it over a network. Each layer adds its own information to ensure proper communication.

2. What is the purpose of the Ethernet header?

   The Ethernet header contains the source MAC address, destination MAC address and Ethertype field. It helps deliver data frames between devices in a local network.

3. What information is stored in the IP header?

   The IP header stores the following information:

- Source IP address
- Destination IP address
- IP version
- Header length
- Total packet length
- Time To Live (TTL)
- Protocol information

4. What is the difference between TCP and UDP headers?

TCP Header	UDP Header
Minimum size is 20 bytes.	Size is 8 bytes.
Connection-oriented.	Connectionless.
Provides reliable data delivery.	Does not guarantee data delivery.
Contains sequence and acknowledgment numbers.	Does not contain sequence and acknowledgment numbers.
Used where reliable communication is important.	Used where low-latency communication is useful.

5. Why does each protocol add a header?

   Each protocol adds a header to provide necessary information for communication, such as addressing, routing, error checking and data delivery. This helps data reach the correct destination.

6. What is packet overhead?

   Packet overhead is the additional information added to the actual application data by protocol headers and other control information during data transmission.
   14+20+20+4=58bytes
   Therefore, the total overhead is 58 bytes.

7. How does header overhead affect network efficiency?
   Header overhead reduces network efficiency because some transmitted bytes are used for protocol information instead of actual application data. Higher overhead means less space is available for useful application data.